Nagar Yuwak Shikshan Sanstha's

Yeshwantrao Chavan College of Engineering

(An Autonomous Institution affiliated to Rashtrasant Tukadoji Maharaj Nagpur University)
Hingna Road, Wanadongri, Nagpur - 441 110
NAAC A++
Ph.: 07104-237919, 234623, 329249, 329250 Fax: 07104-232376, Website: www.ycce.edu

Department of Artificial Intelligence and Data Science

Vision of the Department

To be a preferred hub for nurturing computer engineering aspirants through innovative, ethical, value-based technical education, impactful research, and industry collaboration, to become successful professionals contributing for the betterment of society.

Mission of the Department

To nurture computer engineering professionals by offering:

- *Dynamic learning environment that encourages problem-solving using emerging technology*
- *National Education Policy aligned evidence-based technical education, fostering innovation, research, and industry–institute collaboration*
- *Holistic, ethical and sustainable development for lifelong success.*

Session 2026-2027

Vision: Dream of where you want.	**Mission:** Means to achieve Vision

Program Educational Objectives of the program (PEO): (broad statements that describe the professional and career accomplishments)

PEO1	**Preparation**	**P: Preparation**	**Pep-CL abbreviation pronounce as Pep-si-lL easy to recall**
PEO2	**Core Competence**	**E: Environment (Learning Environment)**	
PEO3	**Breadth**	**P: Professionalism**	
PEO4	**Professionalism**	**C: Core Competence**	
PEO5	**Learning Environment**	**L: Breadth (Learning in diverse areas)**	

Program Outcomes (PO): (statements that describe what a student should be able to do and know by the end of a program)

Keywords of POs:
Engineering knowledge, Problem analysis, Design/development of solutions, Conduct Investigations of Complex Problems, Engineering Tool Usage, The Engineer and The World, Ethics, Individual and Collaborative Team work, Communication, Project Management and Finance, Life-Long Learning

PSO Keywords: Cutting edge technologies, Research

"I am an engineer, and I know how to apply engineering knowledge to investigate, analyse and design solutions to complex problems using tools for entire world following all ethics in a collaborative way with proper management skills throughout my life." *to contribute to the development of cutting-edge technologies and Research*.

Integrity: I will adhere to the Laboratory Code of Conduct and ethics in its entirety.

(Signature and Date in Handwritten)

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Session	**2026-27 (Odd)**	**Course Name**	**LAB: Optimization technique**
Semester	**7th**	**Course Code**	**23ADS1708**
Roll No	**39**	**Name of Student**	**Hrutik Gujarkar**

Practical Number	**6**
Course Outcome	CO1: Understand unconstrained and constrained optimization techniques. CO2: Apply non-linear optimization techniques to solve engineering problems. CO3: Analyze optimization problems using operations research techniques. CO4: Evaluate transportation and transshipment optimization problems.
Aim	To determine the optimum solution using Newton's Optimization Algorithm.
Problem Definition	To find the minimum of the function $f(x, y) = x^4 + 2y^2 - 4x$ using Newton's Optimization Algorithm, starting from the initial point (2, 1) with a tolerance of 10^{-6}, and to determine the optimum point and the optimum value of the function.
Theory (100 words)	Newton's method is a second-order iterative technique for unconstrained optimization. Near the current point X_k, the function is approximated by its quadratic Taylor expansion, using the gradient ∇f and the Hessian matrix H. Minimizing this approximation gives the Newton step $d = -H^{-1}\nabla f$, so the update rule is $X_{k+1} = X_k - H(X_k)^{-1} \nabla f(X_k)$. The method converges quadratically near a minimum where the Hessian is positive definite, but it needs the Hessian to be computed and a reasonable starting point. At the optimum, $\nabla f = 0$ and a positive definite Hessian confirms a minimum.

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<table>
<tr><td></td><td>Step-by-Step Method
1. Define the function f(x, y), the initial point X_0, the tolerance ε and the maximum number of iterations.
2. Compute the gradient ∇f and the Hessian matrix H of the function.
3. At iteration k, evaluate $\nabla f(X_k)$ and $H(X_k)$.
4. If the norm of the gradient is less than ε, stop; X_k is the optimum point.
5. Otherwise, solve $H\,d = -\nabla f$ to obtain the Newton direction d.
6. Update the point: $X_{k+1} = X_k + d$, and repeat from Step 3.
7. At the final point, check that the Hessian is positive definite and calculate f(x, y).</td></tr>
<tr><td rowspan="2">Procedure and Execution (100 Words)</td><td>Algorithm:
1. Define the objective function f(x, y).
2. Compute the gradient and Hessian symbolically.
3. Choose the initial point, tolerance and maximum iterations.
4. Evaluate the gradient and Hessian at the current point.
5. Stop if the norm of the gradient is less than the tolerance.
6. Solve $H\,d = -\nabla f$ to get the Newton step.
7. Update the point X = X + d and repeat.
8. Check the Hessian eigenvalues at the final point.
9. Display the optimum point and the minimum function value.</td></tr>
<tr><td>Code:
<pre>
import numpy as np
import sympy as sp

define variables
x, y = sp.symbols("x y")

Objective function
f = x**4 + 2*y**2 - 4*x

Gradient and Hessian
grad = sp.Matrix([sp.diff(f, x), sp.diff(f, y)])
hess = sp.hessian(f, (x, y))
print("Objective Function:", f)
print("Gradient:", list(grad))
print("Hessian:", hess.tolist())

Convert to numerical functions
f_num = sp.lambdify((x, y), f, "numpy")
g_num = sp.lambdify((x, y), grad, "numpy")
h_num = sp.lambdify((x, y), hess, "numpy")
</pre></td></tr>
</table>

```
# Newton's method
X = np.array([2.0, 1.0])      # initial point
tol = 1e-6
max_iter = 50

print("\nIter        x           y          f(x,y)       ||grad||")
for k in range(max_iter):
    g = np.array(g_num(*X), dtype=float).flatten()
    H = np.array(h_num(*X), dtype=float)
    gn = np.linalg.norm(g)
    print(f"{k:3d}    {X[0]:10.6f} {X[1]:10.6f} "
          f"{f_num(*X):12.6f} {gn:12.6f}")
    if np.linalg.norm(g) < tol:
        print(f"\nConverged in {k} iterations")
        break
    # Newton step: solve H * d = -grad
    d = np.linalg.solve(H, -g)
    X = X + d

# Check second-order condition
eig = np.linalg.eigvals(np.array(h_num(*X), dtype=float))
print("\nOptimal Point: x = %.6f, y = %.6f" % (X[0], X[1]))
print("Minimum Value of f: %.6f" % f_num(*X))
print("Eigenvalues of Hessian:", np.round(eig, 4))
print("Hessian is positive definite:", bool(np.all(eig > 0)))
```

Output:

```
Objective Function: x**4 - 4*x + 2*y**2
Gradient: [4*x**3 - 4, 4*y]
Hessian: [[12*x**2, 0], [0, 4]]

Iter        x           y          f(x,y)       ||grad||
  0     2.000000   1.000000    10.000000    28.284271
  1     1.416667   0.000000    -1.638841     7.372685
  2     1.110534   0.000000    -2.921142     1.478429
  3     1.010637   0.000000    -2.999316     0.129004
  4     1.000112   0.000000    -3.000000     0.001339
  5     1.000000   0.000000    -3.000000     0.000000

Converged in 5 iterations

Optimal Point: x = 1.000000, y = 0.000000
Minimum Value of f: -3.000000
Eigenvalues of Hessian: [12.  4.]
Hessian is positive definite: True
```

Output Analysis	Starting from (2, 1), the algorithm converged in 5 iterations. The gradient norm fell rapidly from 28.28 to 0.129, then 0.0013 and finally 0, which shows the quadratic convergence of Newton's method. The y-value reached 0 in the first step because the y-part of the function is quadratic. The optimum point is (1, 0) with f(1, 0) = −3, which matches the analytical solution of $4x^3 - 4 = 0$ and $4y = 0$. The Hessian eigenvalues are 12 and 4, both positive, so the point is a minimum.

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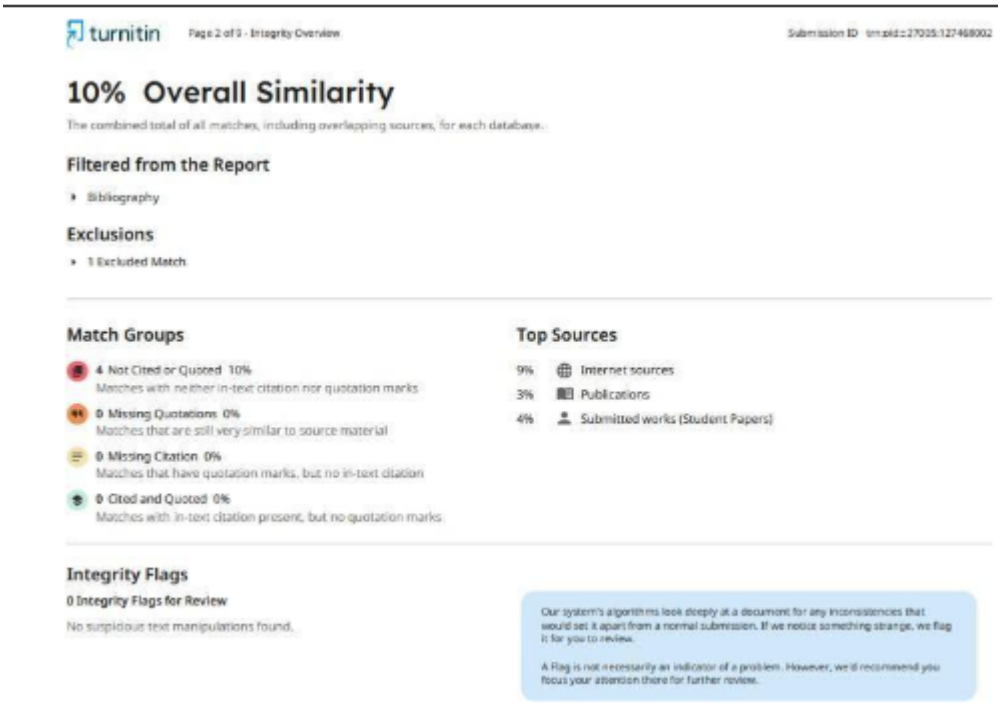
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Link of student Github profile where lab assignment has been uploaded	https://github.com/HrutikGujarkar/Optimizaton-techniques-Practicals
Conclusion	The optimum solution was successfully determined using **Newton's Optimization Algorithm**. The function has its minimum at the point (1, 0), and the minimum value of the function is **−3**.
Plag Report (Similarity index < 12%)	*(Paste plagiarism report screenshot here)* 
Date	___-___-2026